LBP002.03

A Reproducible MCI-Centered Ubiquitin-Proteasome Transcriptional Signature in Peripheral Blood Across Alzheimer's Disease Datasets

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UPS-related transcription shows an unusually strong and reproducible MCI-centered pattern across three blood datasets, driven primarily by MCI-centered reductions and ranking among the strongest GO pathway effects.

Among non-monotonic genes, do UPS genes exhibit larger MCI-centered expression deviations across control, MCI, and AD groups than expression-matched non-UPS genes?

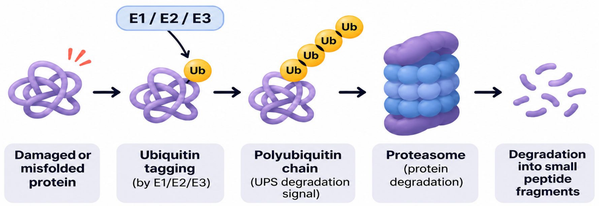
Background

- Alzheimer's Disease (AD) is characterized by abnormal protein accumulation.

- The ubiquitin–proteasome system (UPS) is a major pathway for tagging and degrading damaged or unnecessary proteins.

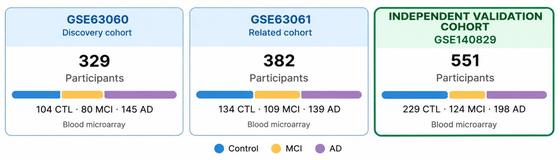
Does whether UPS-related transcription shows a distinctive MCI-centered pattern rather than a simple monotonic change?

The UPS Directs Damaged Proteins for Degradation



The UPS tags damaged or unnecessary proteins with ubiquitin and directs them to the proteasome for degradation.

Three Blood Transcriptomic Cohorts



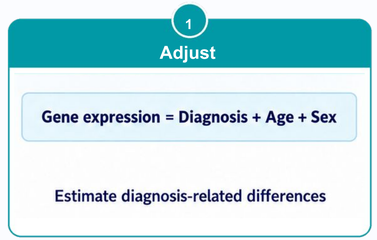
Findings replicated in an independently generated dataset

Genes in Datasets: GO:0043161 - 540 UPS-related genes in the GO set - 391 genes measurable in all three datasets. The analysis included 1,262 participants across control, MCI, and AD groups.

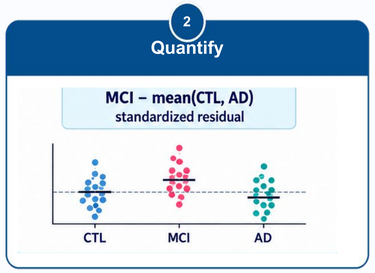
Study Strengths & Robustness

- ✔ Independent validation - Replicated in independently generated GSE140829.
- ✔ Cell-composition sensitivity - Primary UPS enrichment remained significant after adjustment for estimated leukocyte composition.
- ✔ Technical robustness - Results were stable after accounting for relevant batch structure.

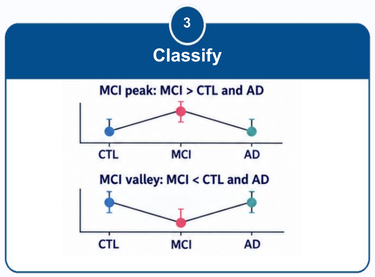
Methodology



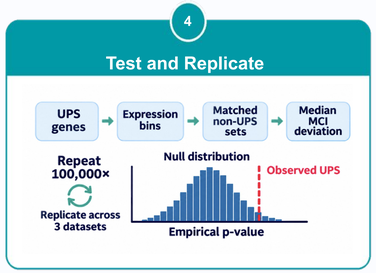
Gene expression was modeled using diagnosis, age, and sex to estimate diagnosis-associated expression differences while accounting for demographic variation.



For each gene, a standardized MCI-centered deviation was calculated to measure how far MCI expression differed from the midpoint between control and AD.

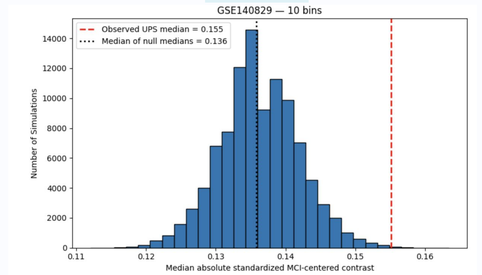


Genes were classified as MCI-centered peaks, valleys, or monotonic patterns based on their age- and sex-adjusted expression across control, MCI, and AD groups.

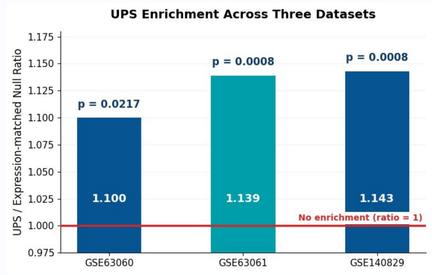


UPS genes were compared with expression-matched non-UPS genes using 100,000-draw Monte Carlo tests, and the same analysis was repeated across independent datasets and comparator pathways.

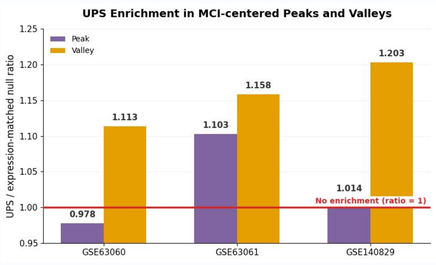
Results



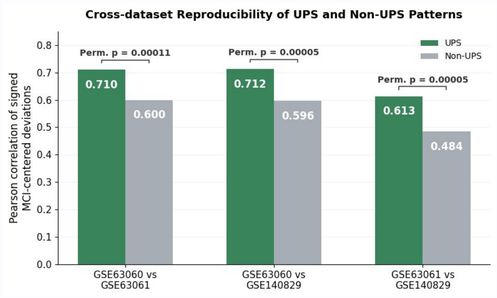
UPS genes show larger MCI-centered deviations than expression-matched non-UPS genes across all three datasets.



UPS genes showed larger MCI-centered deviations than expression-matched non-UPS genes in all three datasets, with significant enrichment in each cohort.

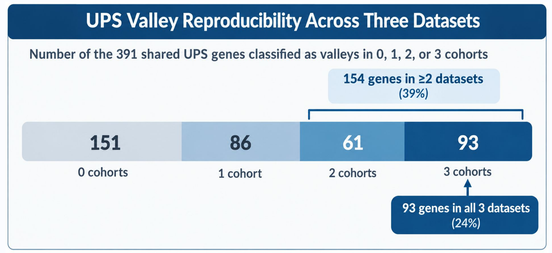


Valley enrichment exceeded peak enrichment in all three datasets, indicating that the UPS signal is driven primarily by MCI-centered reductions rather than increases.



UPS gene-level expression patterns were consistently more reproducible across cohorts than background genes, with significant permutation-test differences in every dataset pair.

A Core Set of UPS Valley Genes Recurs Across Cohorts



Nearly 40% of shared UPS genes were classified as valleys in at least two datasets, including 93 genes (24%) that were valleys in all three cohorts.

UPS Shows Unusually Strong and Reproducible Pathway-Level Enrichment

876 GO pathways tested → **3** significant in all 3 datasets after FDR correction → **2 of 3** strongly overlap UPS

UPS ranked in the top ~5%
by cross-dataset effect magnitude



Only 3 of 876 GO Biological Process pathways were significant in all three datasets, and 2 substantially overlapped with UPS. UPS ranked in the top 5% by cross-dataset effect magnitude.

Limitation & Next Steps

TAKE-HOME
Across three blood cohorts, UPS genes show a reproducible MCI-centered transcriptional deviation, driven primarily by reductions in MCI and ranking among the strongest GO pathway effects.

BIOLOGICAL IMPLICATION
Proteostasis-related transcription may be especially altered around MCI rather than changing monotonically from control to AD.

LIMITATION & NEXT STEP
Because these data are cross-sectional and peripheral, they cannot establish brain mechanism or disease progression. Longitudinal cohorts should test whether the UPS signature tracks within-person cognitive change.

References

UPS graphic: ChatGPT
GO:0043161: MSigDB (2025)
GSE63060/GSE63061: Sood et al., 2015
GSE140829: Nachun et al., 2019